ALAA BOGHDADY

AI Engineer • Generative AI • Machine Learning • NLP

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building enterprise AI solutions using Python, LLMs, RAG, AI agents, OCR, and NLP. Experienced in developing FastAPI AI microservices, bilingual Arabic/English document-intelligence workflows, structured extraction, vector search, and human-in-the-loop AI systems. Strong foundation in Machine Learning, Deep Learning, and Computer Vision, with practical experience integrating AI services with REST APIs and event-driven backend systems.

EXPERIENCE

AI Engineer Trainee | *SustainGRC* May-Jun 2026 (6 weeks)

- **Enterprise AI:** Develop and support AI-powered solutions for regulatory compliance, policy management, document intelligence, and privacy/data-protection workflows.
- **Agentic AI & LLMs:** Build AI-agent workflows using Gemini, prompt engineering, structured outputs, confidence scoring, grounding, validation, and human-review guardrails.
- **RAG & Vector Search:** Implemented semantic obligation matching using sentence-transformer embeddings and Qdrant vector search for policy-management workflows.
- **Document Intelligence:** Developed bilingual Arabic/English PDF processing with PyMuPDF and EasyOCR fallback for extracting obligations, deadlines, recurrence, citations, and compliance metadata.
- **Backend Integration:** Integrated Python/FastAPI AI microservices with backend services through REST APIs and Kafka-based asynchronous workflows; used Git, Bitbucket, Jira, Postman, Docker, and pytest.

NLP Intern / AI Projects | *Cellula Technologies* Sep 2026 - Present

- Built and evaluated RNN/LSTM text-classification models; the selected LSTM achieved ~97% validation accuracy, 0.9398 Macro F1, and 0.9619 Weighted F1.
- Developed a Streamlit demo combining LSTM inference with BLIP image captioning for text and image classification; explored INT8 quantization and memory efficiency.

SELECTED PROJECTS

- **Regulatory Relationship Management AI Service -** 4-agent FastAPI/Gemini system for regulator correspondence ingestion, bilingual reply drafting, obligation extraction, and matter summarization; includes hybrid PDF/OCR routing, confidence thresholds, guardrails, and backend synchronization.
- **Policy Management AI Service -** AI-assisted policy drafting and obligation recommendations using Gemini, sentence-transformers, Qdrant vector search, Kafka signals, and FastAPI endpoints.
- **Privacy & Data Protection AI Layer -** 3-agent workflow for ROPA drafting and privacy-risk proposals with JSON-schema validation, grounding/citations, repair passes, human-in-the-loop controls, and 57 pytest tests.
- **Chest X-ray Pneumonia Detection -** Compared classical ML and CNN/transfer-learning approaches for pneumonia classification from chest X-ray images.

TECHNICAL SKILLS

Programming & Backend: Python, FastAPI, REST APIs, JSON, Pydantic, Kafka, AI Microservices
Generative AI: LLMs, RAG, AI Agents, Prompt Engineering, Gemini, Embeddings, Vector Search, Qdrant, Document Intelligence, OCR
Machine Learning & DL: scikit-learn, TensorFlow, Keras, PyTorch (Basics), Classification, Regression, CNNs, RNNs, LSTMs
NLP & Computer Vision: Text Classification, BLIP, Image Classification, Object Detection, Segmentation
Data & Tools: Pandas, NumPy, Matplotlib, Seaborn, Git, GitHub, Bitbucket, Jira, Postman, Streamlit, pytest, Docker (Basics), AWS/GCP (Basics)

EDUCATION

Bachelor of Commerce - Accounting | Aswan University | 2020 - 2024

CERTIFICATIONS

AI Diploma (INSTANT) • Applications of Artificial Intelligence (BUE) • AI Level 1 (Mindset) • Machine Learning Specialization (DeepLearning.AI - Andrew Ng) • Machine Learning Process A-Z, Statistics & Mathematics (365 Data Science)

LANGUAGES

Arabic - Native | English - Professional Working Proficiency